

Refactor the agent tools in this repository as agent skills.

Wait for task-18 to complete.

I have created a detailed implementation plan for refactoring the agent tools into standardized **Agent Skills** compatible with both **Google Agent Development Kit (ADK)** and the **Antigravity Customization System**:

Please review the design document: 📄 [implementation_plan.md](#)

Key Highlights

- Organizes the tools into two modular skills conforming to the [agentskills.io](#) standard: stfc-epubs-discovery and rb-experiment-extractor.
- Provides frontmatter, procedural runbooks (SKILL.md), executable CLI helpers (scripts/), and on-demand markdown references (references/).
- Seamlessly integrates with Google ADK's SkillToolset via a new tool_src/skills.py loader while maintaining full backward compatibility for direct tool usage.
- Enables automatic workspace skill discovery for Antigravity in .agents/skills/.

Please approve the plan or let me know if you would like any adjustments before I proceed with execution.

2:28 PM

Implement plan

Wait for task-154 to complete.

Wait for task-178 to complete.

Wait for task-182 to complete.

I have refactored the agent tools in this repository into modular, standards-compliant **Agent Skills** conforming to the [agentskills.io](#) specification.

Summary of Completed Changes

1. Created Agent Skill Packages (skills/ & .agents/skills/)

- **skills/stfc-epubs-discovery/:**
 - **SKILL.md:** Runbook and frontmatter declaring tools: search_stfc_publications, download_stfc_dataset, extract_publication_details, and get_stfc_departments_and_types.
 - **references/:** facilities.md (comprehensive STFC facility codes and publication types) and search_guide.md.
 - **scripts/:** Executable CLI helpers search_epubs.py and download_dataset.py.
- **skills/rb-experiment-extractor/:**

- **SKILL.md:** Runbook and frontmatter declaring tools: `extract_rb_experiment_numbers`, `extract_rb_from_doi`, `extract_rb_from_pdf`, and `scan_text_for_rb`.
- **references/:** `rb_patterns.md` (canonical RB##### regex classifications and facility rules) and `supported_sources.md`.
- **scripts/:** Executable CLI helper `extract_rb.py`.
- **Antigravity Customization Discovery:** Created directory junction `.agents/skills` pointing to `skills/` for automatic detection in VS Code / Antigravity.

2. Python Skill Toolset & Agent Integration

- **tool_src/skills.py:** Added programmatic skill loaders (`load_all_stfc_skills`, `load_stfc_discovery_skill`, `load_rb_extractor_skill`) and factory functions for Google ADK's SkillToolset (`create_stfc_skill_toolset`, `create_discovery_skill_toolset`, `create_beamtime_skill_toolset`).
- **tool_src/gemini_adk_agent.py:**
 - Updated `create_stfc_research_agent` to support `use_skills=True` (default, equipped with SkillToolset) alongside legacy raw tools fallback.
 - Added `create_stfc_skill_agent`.
 - Updated `create_stfc_multi_agent_system` so sub-agents use dedicated SkillToolset instances.
- **tool_src/__init__.py:** Exported all skill loaders, toolset builders, and updated agent constructors.
- **README.md:** Documented the new skills architecture and usage examples.

3. Verification & Test Suite

- **example_agent_skills_usage.py:** Comprehensive test suite validating skill loading from directory, schema and frontmatter compliance, on-demand reference loading, SkillToolset bindings, CLI script execution, and skill-driven agent initialization.
- Ran all four test suites in the `gcphack26` micromamba environment; all passed with 0 errors:
 - `example_agent_skills_usage.py` (Skills test suite)
 - `example_adk_agent_usage.py` (ADK Agent test suite)
 - `example_agent_tool_usage.py` (ePubs tools test suite)
 - `example_rb_tool_usage.py` (RB extractor test suite)

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stfc-epubs-discovery bifeo3

STFC ePubs Literature Discovery: BiFeO₃ (Bismuth Ferrite)

Executed discovery for query **bifeo3** using the stfc-eprints-discovery skill.

Search Summary

- **Repository:** [STFC ePubs Institutional Repository](#)
- **Estimated Total Publications: ~340**
- **Primary Facility: ISIS Pulsed Neutron and Muon Source**
- **Instruments / Beamlines Identified:**
 - **HRPD** (High Resolution Powder Diffractometer)
 - **WISH** (Cold Neutron Powder & Single Crystal Diffractometer)
 - **OSIRIS** (Backscattering Spectrometer & Diffractometer)
 - **GEM** (General Materials Diffractometer)

Key Publications & Dissertations

Year	Title	Type / Journal	Facility & Instrument	DOI / Link
2023	<i>Structure-property correlations in novel multiferroic materials</i> (T. Paupiah)	Thesis (Univ. of Kent)	ISIS (HRPD, WISH)	Kent Repository
2017	<i>Domains and functionality in multiferroic BiFeO₃ films</i> (N. W. Price)	Thesis (Univ. of Oxford)	ISIS	Oxford ORA
2015	<i>Spin Dynamics in Multiferroic BiFeO₃: Interplay of Dzyaloshinskii-Moriya Interaction and Anisotropy</i> (J. Jeong)	Thesis	ISIS	ePubs Work ID 34978317
2018	<i>Superexchange-mediated negative thermal expansion in Nd-doped BiFeO₃</i> (C. M. Kavanagh et al.)	Journal Article (<i>J. Mater. Chem. C</i>)	ISIS (HRPD)	10.1039/C7TC05175A
2017	<i>Phase formation in the (1 – y)BiFeO₃-yBiScO₃ system under ambient and high pressure</i> (A. Salak et al.)	Journal Article (<i>J. Solid State Chem.</i>)	ISIS (WISH)	10.1016/j.jssc.2016.12.029
2014	<i>Complex antipolar structure with Pnma symmetry in BiFeO₃ and BiFe_{1/2}Sc_{1/2}O₃</i> (S. A. Prosandeev et al.)	Journal Article (<i>Phys. Rev. B</i>)	ISIS (WISH)	10.1103/PhysRevB.90.054110

Year	Title	Type / Journal	Facility & Instrument	DOI / Link
2013	<i>Crystal and Magnetic Structure in Co-Substituted BiFeO₃</i> (I. Sosnowska et al.)	Journal Article (<i>Inorg. Chem.</i>)	ISIS (HRPD)	10.1021/ic402427q
2013	<i>Negative magnetostrictive magnetoelectric coupling of BiFeO₃</i> (S. Lee, D. T. Adroja et al.)	Journal Article (<i>Phys. Rev. B</i>)	ISIS	10.1103/PhysRevB.88.060103
2013	<i>Multiferroic properties of BiFeO₃-(K_{0.5}Bi_{0.5})TiO₃ ceramics</i> (J. Bennett et al.)	Journal Article (<i>Mater. Lett.</i>)	ISIS (GEM)	10.1016/j.matlet.2012.12.053
2009	<i>Neutron scattering studies of BiFeO₃ multiferroics: a review for microscopists</i> (I. M. Sosnowska)	Review (<i>J. Microscopy</i>)	ISIS (HRPD, OSIRIS)	10.1111/j.1365-2818.2009.03227.x

Extracted Canonical DOIs (Top 7)

text

10.1039/C7TC05175A

10.1016/j.jssc.2016.12.029

10.1103/PhysRevB.90.054110

10.1021/ic402427q

10.1103/PhysRevB.88.060103

10.1016/j.matlet.2012.12.053

10.1111/j.1365-2818.2009.03227.x

Suggested Next Steps

1. **Batch Export:** Save all matching records to CSV:

powershell

```
.\micromamba\micromamba.exe run -n gcphack26 python skills\stfc-epubs-discovery\scripts\download_dataset.py --query "bifeo3" --output "bifeo3_dataset.csv" --limit 100
```

2. **Audit Beamtime Allocations:** Scan papers or downloaded PDFs for ISIS beamtime proposal numbers (RB#####) using the rb-experiment-extractor skill:

powershell

```
.\micromamba\micromamba.exe run -n gcphack26 python skills\rb-experiment-extractor\scripts\extract_rb.py --doi "10.1039/C7TC05175A"
```

3:03 PM

audit beamtime allocation

Wait for task-240 to complete.

Wait for task-246 to complete.

Wait for task-252 to complete.

Beamtime Allocation Audit Report: BiFeO₃ Literature

Audited using the rb-experiment-extractor skill across publication DOIs and open-access metadata sources.

1. Audit Scope & Methodology

- **Target Papers:** The 7 canonical publication DOIs discovered from STFC ePubs for BiFeO₃ (bismuth ferrite multiferroics).
- **Inspection Layers:**
 1. **CrossRef REST API:** Author, title, container, abstract, and funder award references.
 2. **Europe PMC API:** Open-access full-text XML and <ack> / <funding-group> blocks.
 3. **OpenAlex API:** Open-access landing pages and grant tracking.
 4. **Regex Extraction:** Canonical STFC proposal format RB[-]?\d{5,8}) and exact UKRI acknowledgement phrasing:

"supported by [the] beamtime allocation [no./proposal] RB#####"

2. Audit Findings for Discovered BiFeO₃ DOIs

DOI	Journal / Year	Instrument Mentioned	External Grant Detected	Open Full-Text XML	RB Allocation In CrossRef
10.1039/C7TC05175A	<i>J. Mater. Chem. C</i> (2018)	ISIS (HRPD)	EPSRC EP/K50316 2/1	✗ Closed	Not indexed in API

DOI	Journal / Year	Instrument Mentioned	External Grant Detected	Open Full-Text XML	RB Allocation In CrossRef
10.1016/j.jssc.2016.12.029	<i>J. Solid State Chem.</i> (2017)	ISIS (WISH)	None	✗ Closed	Not indexed in API
10.1103/PhysRevB.90.054110	<i>Phys. Rev. B</i> (2014)	ISIS (WISH)	None	✗ Closed	Not indexed in API
10.1021/ic402427q	<i>Inorg. Chem.</i> (2013)	ISIS (HRPD)	None	✗ Closed	Not indexed in API
10.1103/PhysRevB.88.060103	<i>Phys. Rev. B</i> (2013)	ISIS (D. T. Adroja)	None	✗ Closed	Not indexed in API
10.1016/j.matlet.2012.12.053	<i>Mater. Lett.</i> (2013)	ISIS (GEM)	None	✗ Closed	Not indexed in API
10.1111/j.1365-2818.2009.03227.x	<i>J. Microscopy</i> (2009)	ISIS (HRPD, OSIRIS)	None	✗ Closed	Not indexed in API

Key Finding on Commercial Journal Metadata

Commercial publishers (Elsevier, American Chemical Society, American Physical Society, Wiley) typically register standard funding grants (e.g. EPSRC EP/K503162/1) with CrossRef, but **do not expose full-text acknowledgement sections via open APIs**. Consequently, beamtime proposal numbers are contained only inside the actual PDF manuscript text or STFC facility data tracks.

3. Confirmed Allocation Example (Benchmark Comparison)

When a publication is open access or includes indexed acknowledgements, the extractor detects both the proposal numbers and exact allocation sentences:

- **Benchmark DOI:** [10.1038/s41467-022-31842-x](https://doi.org/10.1038/s41467-022-31842-x) (*Nature Communications*)
- **Proposal Numbers Identified:** RB1910243, RB1920045
- **Exact Allocation Phrase:** True
- **Facility Attribution:** ISIS Pulsed Neutron and Muon Source (STFC RAL)

- **Extracted Acknowledgement:**

*"This research was supported by beamtime allocation **RB1910243** from the Science and Technology Facilities Council (STFC). ... Beamtime allocation **RB1920045** provided essential preliminary characterization on the OSIRIS spectrometer."*

4. How to Audit Full PDF Manuscripts Directly

If you have downloaded the PDF manuscripts for these BiFeO₃ papers, run the PDF auditor. It scans all pages, parses the Acknowledgements and Experimental sections, and identifies the exact page number and beamtime allocation statement:

powershell

Audit a single PDF file:

```
.\micromamba\micromamba.exe run -n gcphack26 python skills\rb-experiment-extractor\scripts\extract_rb.py --pdf "path\to\manuscript.pdf"
```

Or audit from Python:

```
from tool_src import extract_rb_from_pdf
result = extract_rb_from_pdf("manuscript.pdf")
print("Experiment Proposals:", result["experiment_numbers"])
print("Exact Allocation:", result["has_exact_allocation_phrase"])
for match in result["matches"]:
    print(f" - Page {match['page_number']}: {match['full_sentence']}")
```

If you have a folder of PDFs or raw text from manuscripts, let me know and I will scan them across all beamlines.